Final Technical Report

Boréal Marché Customer Intelligence Platform

Manish Chauhan (ID 6401170) · Individual project · 420-977-VA AI Development Project, Sect. 91401 · Vanier College · Instructor: Steve Ataky, Ph.D. (Eng.) · 26 October 2026

Consolidates D2 (design), D3 (models), D4 (deployment) and D5 (optimisation and ethics). Repository: github.com/AI-DS-Git/boreal-marche-customer-intelligence. Every number in this report comes from a script in the repository or from a measurement on the live service; where something was estimated or not measured, the text says so.

1. Summary

Boréal Marché, a Québec retailer, sends the same coupon to every customer. I built a platform with three AI components and operated it as a real service: **Head A** estimates which customers will not buy again in the next 90 days; **Head B** recommends products for a basket; **Head C** is a bilingual assistant that answers policy and product questions with sources. The system runs on AWS in Canada behind HTTPS, is tested and deployed automatically, watches its own inputs, and is documented against Québec's Law 25. The data are a public retail dataset (UCI Online Retail II) presented as Boréal Marché's; no real customer data was used.

Question	Result	Compared with	Sec.
Does the retention model beat simple alternatives?	Held-out PR-AUC 0.8753, ROC-AUC 0.7301 (788 customers, evaluated once)	Cross-validated PR-AUC: dummy 0.673, one-rule heuristic 0.710, Logistic Regression 0.799, Extra Trees (untuned) 0.819	4
Is it worth acting on?	Expected profit $1,067 on 788 held-out customers with targeted offers	Blanket coupon $235; no action $0. Projection about $22,000 a year (planning estimate, assumptions in Section 4.5)	4.5
Do recommendations help?	Hit-rate@5 0.404, catalogue coverage 3.15%	Popularity baseline 0.312 and 0.32%	5
Is the assistant safe?	Golden set (40 questions, EN/FR): tool choice 97.5%, refusals 97.5%, citations 95%	In-sample (the evidence floor was set on the same questions); real LLMs not measured	6
Is it fast?	Live POST /predict median 109 ms, p95 141 ms, from a laptop over the internet	265 / 308 ms before the D5 change	9
Is it fair?	Not for every group: 6 of 10 audited groups are flagged; all customers in the three lowest spend quartiles and the newest tenure band receive an offer review	Four mitigations tested; two did not work	10
Is it lawful and governed?	Human approval before every offer; bilingual notice; 120-day retention; model, data and governance files	Law 25 s.12.1 and s.8.1, primary sources read 26 Oct 2026; not legal advice	11

Three limits shape everything that follows. The results come from a public dataset with a single 788-customer held-out slice. The first real accuracy signal on live predictions cannot exist until 90 days after the first prediction (about 24 January 2027). And the model cannot tell retained from churned customers among new and low-spend customers, which is both a fairness finding and a business limit.

2. Problem, scope and evidence

Business problem. A blanket quarterly coupon costs money on customers who would have ordered anyway and does nothing special for those who are drifting away. Customer questions arrive in English and French and are answered by hand. The platform's decision is narrow: *which customers should a staff member consider for a retention offer*, plus two supporting services. It is decision support, not an autonomous decision-maker.

Data. The UCI Online Retail II workbook (two sheets, 1,067,371 rows, 1 Dec 2009 to 9 Dec 2011, SHA-256 recorded in D2). Cleaning is sequential and counted: exact duplicates (34,335), cancellations (19,104), non-positive quantity (3,393), non-positive price (2,626) leave 1,007,913 rows (94.43%). The workbook is not committed to the repository.

Stage	Rows removed	Rows remaining
Bronze: both sheets loaded	0	1,067,371
Silver 1: exact duplicates	34,335	1,033,036
Silver 2: cancellation invoices	19,104	1,013,932
Silver 3: quantity ≤ 0	3,393	1,010,539
Silver 4: price ≤ 0	2,626	1,007,913
Silver 5: invalid timestamp	0	1,007,913

Table 1. Cleaning. Counts are sequential because the problems overlap.

Label and windows. Features use transactions on or before the cutoff of 31 Aug 2011; the label *retained_90d* is 1 if the customer orders at least once in the next 90 days (to 29 Nov 2011). The Gold table has 5,249 eligible customers and 42.7% positives; the chronological held-out slice (the last 15%, 788 customers) has a higher retained share (76%) because it holds the customers who purchased most recently. Train, calibration and held-out slices are 70/15/15% by last-purchase date; the held-out slice was used once, after every modelling decision was frozen.

Leakage was tested, not assumed. Three checks: recency and monetary features recomputed from raw rows before the cutoff matched the shipped table exactly; deliberately injecting two label-window aggregates pushed PR-AUC from 0.857 to 1.000 (the signature that would have flagged leakage); and the highest feature-label correlation was 0.431 (recency), below the 0.85 investigation trigger.

3. Architecture and key decisions

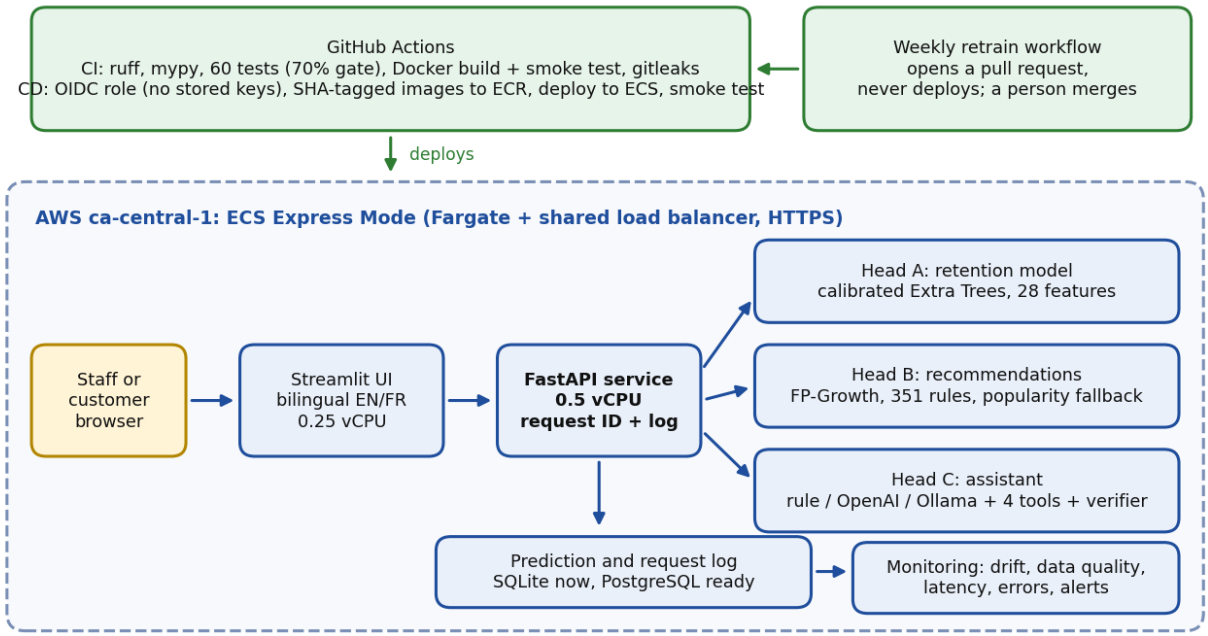


Figure 1. Runtime and delivery architecture. Models are loaded once at start-up; the log feeds the monitoring page; GitHub Actions deploys through OIDC, with no stored AWS keys.

Decision	Why	Trade-off
One joblib artifact bundles preprocessing, Extra Trees and the isotonic calibrator	Serving cannot apply different preprocessing than training; a bad artifact fails the container before /ready is green	joblib loads by unpickling: only trusted artifacts; library versions pinned to the metadata (scikit-learn 1.8.0)
Calibrated probabilities and a profit-based threshold (0.82)	The decision is a cost trade-off, so probabilities must mean something	Isotonic calibration is a step function: scores tie, and are capped near 0.52 for low-spend and new customers
Association rules (FP-Growth) instead of collaborative filtering	Baskets, no ratings; explainable ("because you added X"); microsecond lookup	Catalogue coverage is only 3.15%; popularity fallback
Assistant: retrieval plus four typed tools plus a verifier, not fine-tuning	Policies change without retraining; answers can cite; claims can be checked	Real-LLM quality not yet measured
ECS Express Mode (Fargate + ALB) instead of App Runner	App Runner is closed to new customers and Streamlit needs WebSockets	Shared ALB is the main fixed cost; one task per service
Human approval before every offer	Law 25 s.12.1 duties attach to exclusively automated decisions	Staff time; not a rubber stamp: reviewers see the factor groups

Table 2. Decisions that shaped the design.

4. Head A: retention model

4.1 Features. 28 predictors computed from pre-cutoff transactions, in eight groups (Table 3a). D2 had 22; six were added in D3 to meet the 25+ requirement. Two of them can be undefined (a customer with one order has no gap between orders; 1,580 of 5,249) and are median-imputed inside the pipeline, not zero-filled.

Group	Predictors
Recency	recency_days, last_purchase_date (as days)
Order frequency	frequency_orders, active_days, active_months, orders_per_active_month, avg_days_between_orders
Spend and volume	monetary_total, monetary_mean_order, revenue_per_order, total_quantity, average_quantity, quantity_per_order, transaction_count
Basket variety	unique_products, unique_product_descriptions, products_per_order, avg_lines_per_order
Price level	average_unit_price, price_std
Tenure	customer_lifetime_days, first_purchase_date (as days)
Seasonality and weekends	weekend_transactions, weekend_transaction_ratio, holiday_transactions, holiday_season_ratio
Country	country, is_uk

Table 3a. The 28 predictors by group (the same groups are used for importance and explanations in Section 10).

4.2 Model comparison. Nine model families and two baselines under identical three-fold chronological cross-validation. Gaussian Naive Bayes is indistinguishable from the dummy (PR-AUC 0.675, ROC-AUC 0.505): its independence assumption is badly violated by these correlated features. It is reported, not dropped.

Model	Mean PR-AUC	Std	Mean ROC-AUC	Brier
Extra Trees	0.8189	0.067	0.7073	0.197
Random Forest	0.8080	0.068	0.6985	0.195
Hist Gradient Boosting	0.8034	0.069	0.6835	0.210
Logistic Regression	0.7985	0.064	0.6707	0.218
AdaBoost	0.7946	0.082	0.6877	0.219
K-Nearest Neighbours	0.7901	0.069	0.6899	0.213
Gradient Boosting	0.7876	0.094	0.6693	0.218
Decision Tree	0.7811	0.081	0.6675	0.213
Business rule (recency ≤ 45 days)	0.7095	0.048	0.5552	0.272
Gaussian Naive Bayes	0.6753	0.070	0.5054	0.333
Dummy (prior)	0.6730	0.069	0.5000	0.351

Table 3. Spot-check, three chronological folds. XGBoost and LightGBM were not available in the build sandbox, so scikit-learn's HistGradientBoosting stands in; this is stated, not hidden.

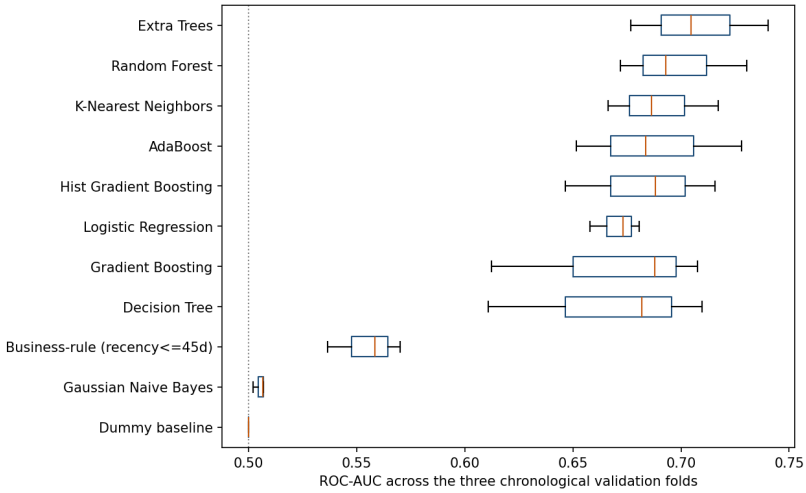


Figure 2. ROC-AUC across folds for all candidates and baselines.

4.3 Tuning. A 60-trial Optuna TPE search per model with a median pruner over the same folds, logged to MLflow (122 runs). Extra Trees improved from 0.8189 to 0.8347 (n_estimators 450, max_depth 15, min_samples_leaf 27, min_samples_split 30, all features); Random Forest from 0.8080 to 0.8333 (a much shallower forest, depth 3). Depth and leaf size were the dominant levers; trials with small leaves or unlimited depth were pruned early. I ran the search on my own machine; the managed SageMaker tuning job is written and tested in script mode but **was not run**, so there is no managed-training comparison in this report.

4.4 Calibration, threshold and held-out test. Isotonic regression was chosen over Platt scaling because forest scores are not sigmoid-shaped. On the held-out slice Extra Trees gives PR-AUC 0.8753, ROC-AUC 0.7301, Brier 0.1630. Random Forest scored slightly higher (0.8787); I did not switch, because choosing after seeing the held-out result would defeat its purpose. The threshold is set by expected profit: a customer is a candidate for an offer if P(retain) is below 0.82. **A methodological note:** D3 swept the threshold on the held-out slice, so the held-out profit was slightly favoured by that choice. Choosing it on the calibration slice instead (scripts/profit_sensitivity.py) gives 0.825, the same point on the grid, and the same held-out profit ($1,067), so the result does not depend on it; the calibration scores are partly in-sample for the calibrator, so this is a consistency check and not a clean test.

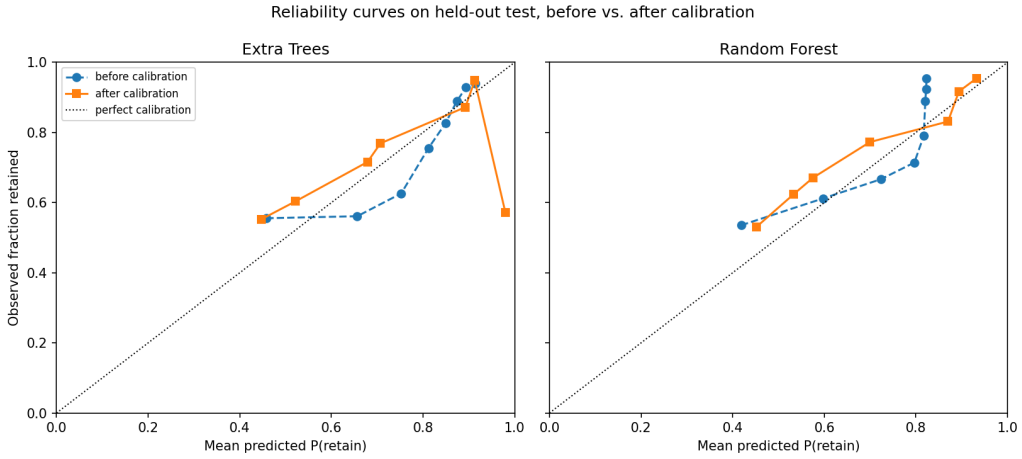


Figure 3. Reliability curves on the held-out slice, before and after calibration.

4.5 Business value and how fragile it is. Planning assumptions (not Boréal Marché financials): coupon $5, gross margin 20%, a coupon saves 30% of true churners. With the deployed model and the frozen threshold, scripts/profit_sensitivity.py gives, on the 788 held-out customers, targeted profit $1,067 against $235 for the blanket coupon and $0 for no action: a gain of $1.06 per customer per campaign, about $22,000 a year for 5,249 customers contacted four times (D3 reported $1,066 against $201 on its own split; the difference is in the blanket baseline and I did not trace it further). Changing one assumption at a time the gain stays positive (a $3 coupon: $2,900 a year; $8:

$51,000; recovery 45%: $9,100; recovery 10%: $39,500; margin 25%: $15,600; margin 15%: $28,700). **Changing all three together it does not:** in 5 of 27 combinations the targeted policy loses to the blanket coupon (a cheap coupon with a high margin and a high recovery rate, where sending to everyone is simply better). The case for targeting depends on coupons being costly relative to the margin they protect, which is why Section 12 recommends a pilot with a control group.

4.6 Error analysis. By spend tercile, recall for the retained class is 1.9% (low), 53.5% (mid) and 87.9% (high); by country 54.2% (UK) and 50.0% (non-UK). The model has learned that spend is most of the signal; below it, it barely discriminates. This finding carried into the fairness audit (Section 10).

5. Head B: recommendations

Rules are mined from invoice baskets on or before the cutoff only (29,850 baskets, 4,415 products), so the temporal evaluation cannot see the customers' later purchases. The sparse (CSR) matrix uses 3.13 MB against 131.79 MB dense (97.6% less). Apriori (demonstration) and FP-Growth (production path) find identical itemsets: 887 up to size three, in 8.15 s and 5.65 s (1.44x faster). A support/confidence sweep chose 1% support and 30% confidence (351 rules); lower support roughly triples itemsets with mostly noisy long-tail pairs, higher support keeps only the best sellers. Lift filtering is demonstrated on a real pair (21790 → 85123A: confidence 15.9% but lift 1.10, because 85123A is bought by about 14% of customers whatever else is in the basket). The serving engine indexes rules by antecedent, drops items already in the basket, keeps the highest-lift rule per product, returns the firing rule so the interface can say why, and blends in popularity when too few rules fire.

Strategy	Hit-rate@5	Catalogue coverage	Eligible customers
Rule engine	0.4043 (906 / 2,241)	3.15% (139 / 4,415)	2,241
Popularity baseline	0.3119 (699 / 2,241)	0.32% (14 / 4,415)	2,241

Table 4. Temporal hold-out: recommendations from each customer's last pre-cutoff basket, scored on purchases in the next 90 days. The rule engine is 1.30x the popularity baseline with about ten times the coverage.

6. Head C: assistant

A bilingual assistant answers policy and product questions, suggests products and redirects retention questions to the Prediction page. Three providers sit behind one interface: a deterministic **rule** provider (the only one live), and **OpenAI** and **Ollama** providers with tool calling and streaming, implemented and tested against a local fake server but not enabled on the deployed service (no key). Retrieval matches the question to policy chunks by cosine similarity with an evidence floor; below it the assistant refuses and hands off to staff. The deployed retrieval is a hashing embedder, which is lexical, not semantic.

Guardrails sit around the model, not inside it: privacy and out-of-scope requests are refused before any model is called; only four typed tools can supply policy text, product codes and recommendations; and a verifier replaces any model answer containing a product code or number that no tool returned. A 40-question golden set (15 policy, 10 tool-call, 8 out-of-scope, 7 French) is scored offline in CI: tool choice 97.5%, refusals 97.5%, citations 95%. **These figures are in-sample**: the evidence floor was set on the same questions. Real-model tool choice and real-embedding retrieval quality have not been measured (scripts/eval_assistant.py exists and has not been run).

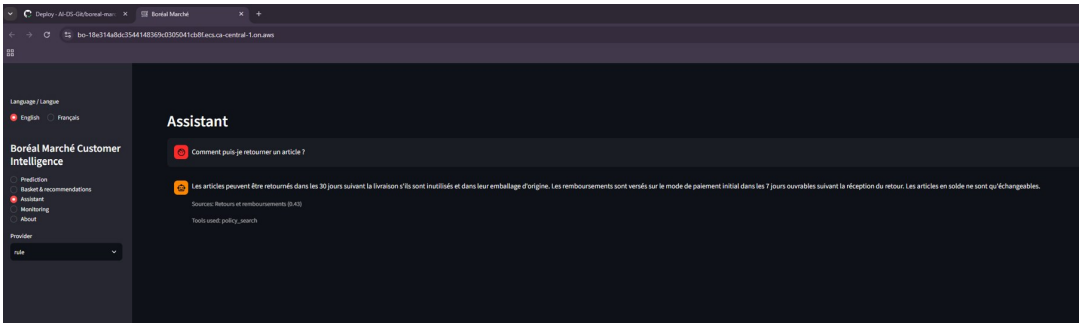


Figure 4. The live assistant answers a French question in French, citing its source and the tool used.

7. Deployment, CI and CD

The API (FastAPI, Pydantic v2 with unknown fields rejected) and the bilingual Streamlit interface run as two containers on AWS ECS Express Mode in ca-central-1 (API 0.5 vCPU / 1 GB, UI 0.25 vCPU / 0.5 GB, one task each), behind HTTPS. Endpoints: /health (liveness), /ready (model loaded), /version (commit SHA, model version, threshold), /predict, /predict/batch, /explain, /recommend, /chat, /chat/stream, /monitoring/summary, /monitoring/outcomes, /metrics. Every response carries a request ID; errors are uniform JSON. A golden-prediction test fixes customer 12347 at 0.8095, so a silent change of artifact or preprocessing fails the build. Images are multi-stage, non-root, with a health check; scikit-learn, pandas, numpy and joblib are pinned to the versions in the model metadata.

Stage	What it enforces
CI: lint	ruff (E, F, W, B) and mypy
CI: test	pytest, 70% coverage gate: golden prediction, rules, assistant, monitoring maths, data quality, API contract (60 tests)
CI: docker	Builds both images, starts the API image, checks /ready and that /version equals the commit SHA
CI: secret-scan	gitleaks over the whole history
CD	Runs only after CI succeeds on main; GitHub OIDC to a role pinned to this repository and branch (no stored AWS keys); images tagged with the commit SHA; waits until the live /version reports the new SHA; smoke test; about 20 minutes
Branch protection	Pull request required; the four checks required; force-push and deletion disallowed (approvals off: single developer)

Table 5. Pipeline. Rollback is re-running Deploy on a previous SHA.

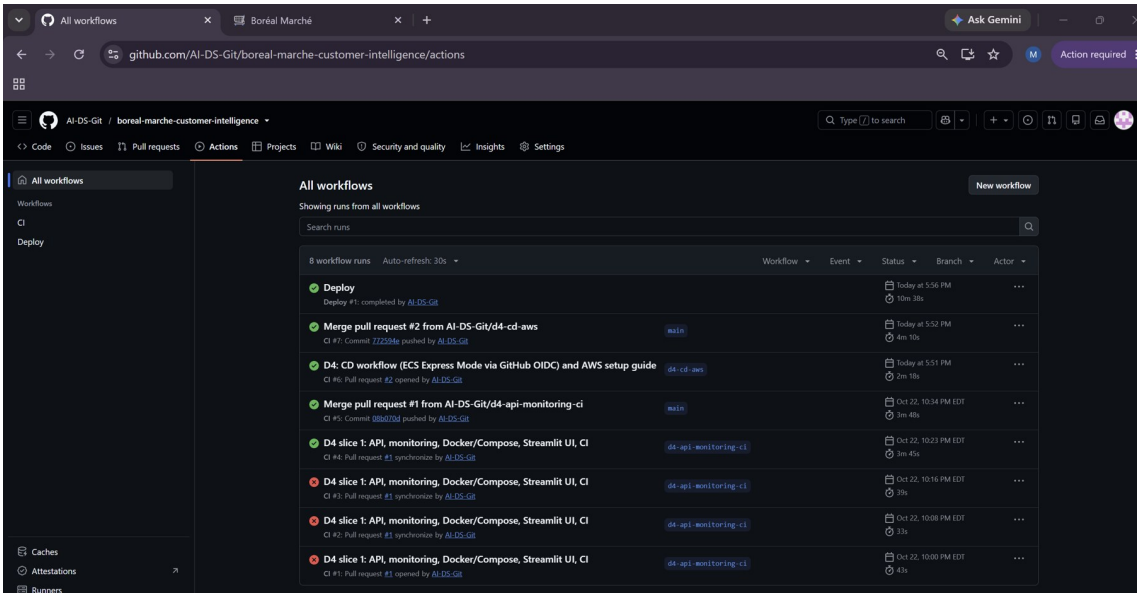


Figure 5. Actions history during D4: CI red then green, merges, and the first deploys.

8. Monitoring and retraining policy

The monitoring page and /monitoring/summary compare live inputs with the training reference. Input and score drift use the Population Stability Index with a sampling-noise correction (amber at 0.10, red at 0.25; the KS statistic is shown for context only); data quality tracks out-of-range share and null-rate change per feature; operations track volume, p50/p95/p99 latency and the 5xx share (red at 5% and at p95 ≥ 500 ms). Fewer than 30 predictions shows "insufficient data", because small samples show amber from noise alone (seen at n of about 320). Only red drift or operations states raise an alert (de-duplicated, 6-hour cooldown, optional webhook).

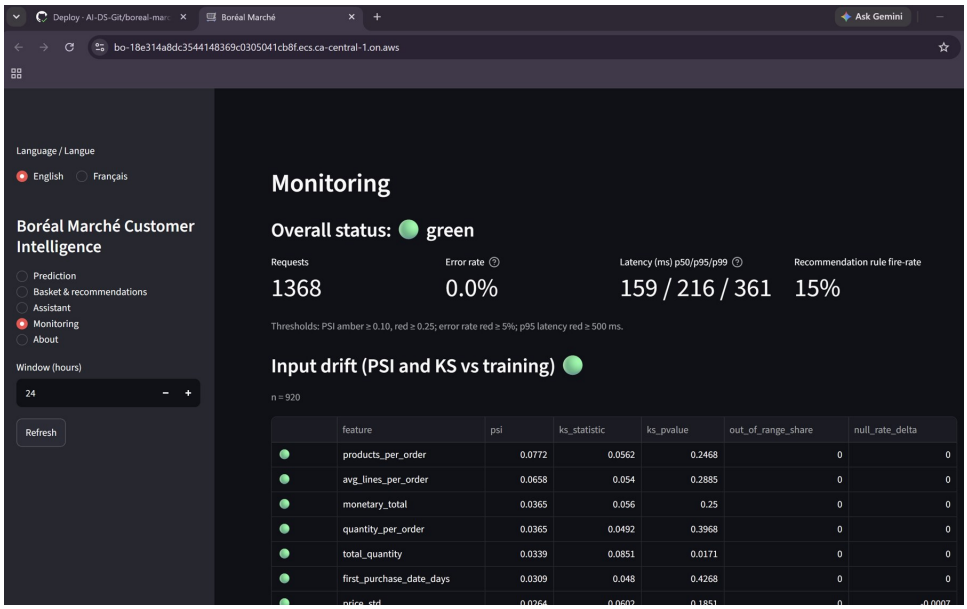


Figure 6. Live monitoring page after normal traffic: all PSI under 0.10, green, 0.0% errors.

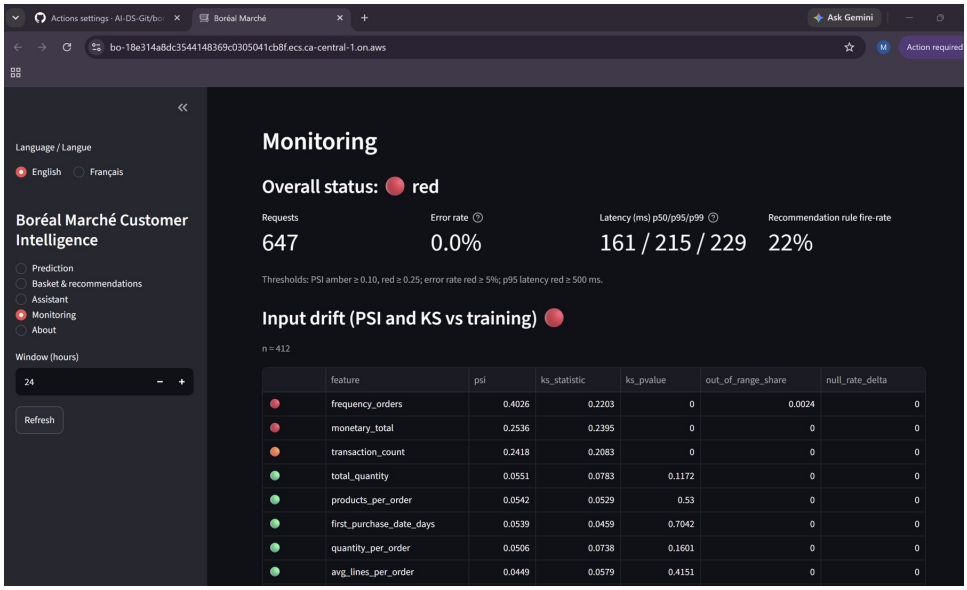


Figure 7. The same page after deliberately drifted traffic: frequency and monetary features turn red.

Delayed ground truth. Retention is observable only 90 days after a prediction. Each logged prediction keeps its request ID; true labels are posted later to /monitoring/outcomes and joined. The page states the date of the first true accuracy signal (about 24 Jan 2027) and the number of labelled predictions; until then only drift and operations signals are live, and the page says so.

Retraining rule. A weekly workflow retrains the frozen recipe and scores candidate and deployed model on the same held-out slice. It opens a pull request only if PR-AUC improves by at least 0.005, ROC-AUC does not fall by more than 0.01, Brier does not worsen by more than 0.005, the schema is unchanged and the data has not shrunk by more than 10%. It never deploys: a person merges, CI runs, then Deploy. The dataset is static, so this demonstrates the mechanism and not real improvement; the first run kept the current model (0.8782 against 0.8764). Proposed but not automated: a review after three days of red drift or when labelled PR-AUC is more than 0.05 below the held-out value, and a fairness check in the promotion gate (Section 10).

9. Performance, scalability and cost

Latency. The saved forest has 450 trees and n_jobs=-1, so scikit-learn started a thread pool on every call. Serving it single-threaded gave identical outputs (the golden test still passes) and was faster at every size on the development machine: a single prediction fell from 110.5 ms to 35.0 ms, a batch of 1,000 from 131 ms to 46 ms. Loading the model per request would cost about 65 ms more than loading it once at start-up. Explanations score all factor groups in one call (39 ms against 286 ms for one call per group).

Live endpoint (laptop, 60 sequential requests)	Before p50	Before p95	After p50	After p95
GET /health	52.3	72.1	63.0	81.0
POST /predict	265.3	307.5	109.5	141.2
POST /explain	not deployed	-	229.8	329.0
POST /predict/batch (100)	972.9	1,795.1	826.4	934.0
POST /recommend	67.9	86.0	68.9	84.7
POST /chat (rule provider)	65.4	78.5	71.2	87.0

Table 6. Live latency in ms, 26 Oct 2026, before and after the D5 change (merge commit eddef8f). The 50–65 ms round trip is included in every figure; a prediction cost the server roughly 210 ms before and 45 ms after (an estimate). Throughput under concurrent load was not measured.

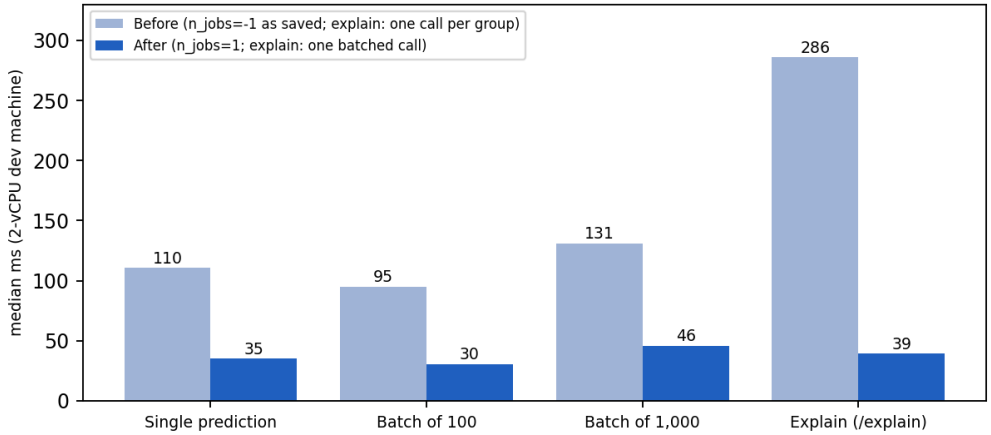


Figure 8. In-process before and after for the three optimisations (development machine).

Where it breaks first. One API task handles an estimated 12–14 requests/s (not load-tested on Fargate). The container-local SQLite log breaks next (about 500 writes/s measured, not shared between tasks, wiped at every redeploy), then the monitoring summary (about 1 s of CPU per refresh at 5,000 logged rows, which is also the cap). Rule lookups stay in microseconds to about 100,000 rules; retraining memory becomes the limit near a million customers. First change: move the log to PostgreSQL, then add tasks.

Cost. Always-on cost is about $0.06 an hour, roughly $45–46 a month (a lower bound: Fargate rates are those of US East, the Canada rate is higher, the load balancer is an assumption). AWS showed $5.09 for October 2026 to date on 26 October (it says credits may cover it). Per 1,000 predictions: $1.50 at 1,000 a day, $0.15 at 10,000, $0.015 at 100,000 for the fixed stack; about $0.0004 marginal compute. Per 1,000 assistant queries: about $0 for the rule provider; about $0.26 for OpenAI gpt-4o-mini, estimated from prompt sizes at list prices (no key is configured, so no real usage exists); Ollama has no per-token fee but needs a host. The largest lever is switching the stack off when it is not in use. After grading I will delete the ECS services and ECR repositories.

10. Interpretability and fairness

Which inputs matter. Permutation importance by feature group on the held-out slice (15 shuffles): order frequency by far (PR-AUC falls 0.044), then spend and volume (0.008), tenure (0.005), seasonality (0.004); basket variety, price level and country are indistinguishable from zero. **Per customer**, /explain replaces each group with a typical-customer value, re-scores, and returns the largest effects, in uncalibrated score points. A first version used calibrated scores and returned identical effects for four groups because of the calibrator's steps; it now uses the raw forest score. Staff see the top three factors next to the score. They are not shown to customers: they describe how the model reacts, not why a customer behaves as they do; local and global views can disagree (recency leads locally for customer 12347 but has no global importance on this slice); and a group such as country must never appear as a reason.

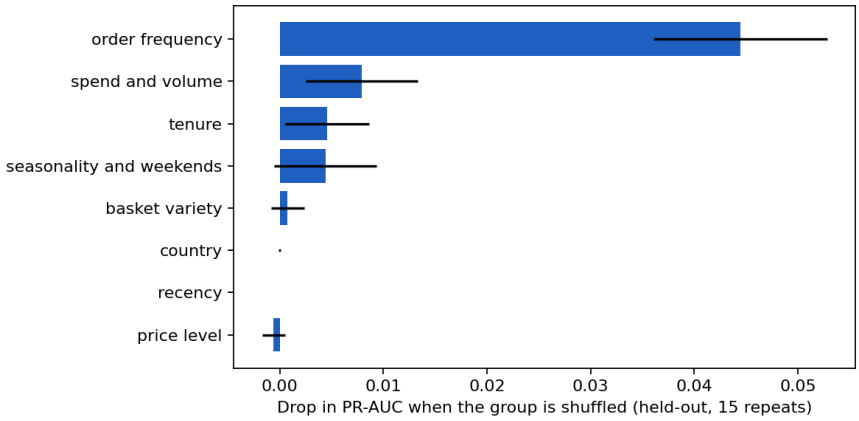


Figure 9. Global importance by feature group.

Fairness. On the 788 held-out customers at threshold 0.82, a group is flagged if its recall for retained customers is more than 5 points below the overall 55%. There are no protected attributes in the data, so this is a behavioural audit, not a legal finding. Six of ten groups are flagged. Spend quartiles Q1–Q3 and customers with 0–90 days of tenure have recall 0%, because the model never scores them above 0.52 while the cut-off is 0.82: every one of them is sent for offer review, although 41–66% of them were retained, so about half of those offers are unnecessary. The 91–365-day tenure band is also flagged (recall 21%). Europe (non-UK) is 9 points below but has only 78 customers and a wide interval, so I treat it as unproven; rest of world has five customers and cannot be interpreted.

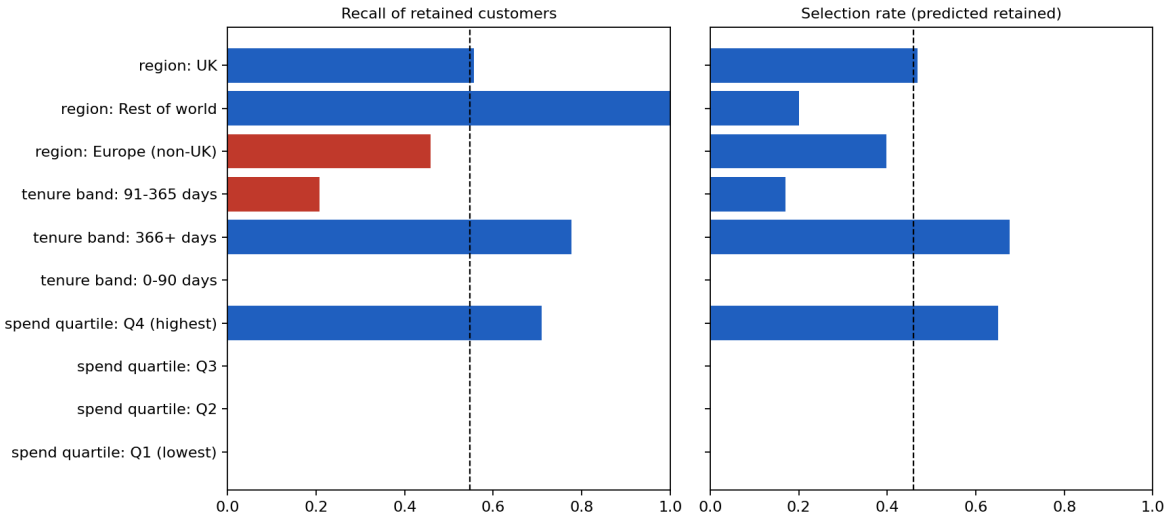


Figure 10. Recall and selection rate by group; flagged groups in red; dashed line = overall.

Mitigation	Recall	Precision	Worst-group recall	Flagged	Verdict
Baseline (single threshold 0.82)	55%	91%	0%	6	Current
A. Group thresholds by spend quartile	68%	83%	38%	4	Pilot only: precision falls; thresholds rest on small groups
B. Rank within group, same review budget	51%	84%	31%	4	Use to order the reviewer queue
C. Per-group recalibration	56%	91%	0%	6	Negative result: calibration is not the cause
D. Retrain with inverse-group weights	56%	90%	0%	6	Negative result: no information to learn from

Table 7. Mitigations on the same held-out slice. The fix that would help is information (for example the first-order basket or acquisition channel), not tuning.

11. Governance under Law 25

I read the Act respecting the protection of personal information in the private sector (CQLR c. P-39.1) on Légis Québec (s.12.1, s.8.1, s.23) and the Commission d'accès à l'information page on automated decisions on 26 October 2026. Section 12.1 requires that a person be told when a decision is based exclusively on automated processing; that, on request, they be told which information was used, the reasons and the main factors, and of their right to correction; and that they can present observations to someone able to review the decision. This is not legal advice.

- **Design consequence.** A human approves every retention offer, and this is why that step exists. The API says target_with_offer, the interface says a staff member reviews it, and reviewers see the factor groups and can override. The score is also profiling (s.8.1) and is disclosed.
- **Notice.** A plain-language bilingual notice is on the application's About page and in GOVERNANCE_LOI25.md; the register covers four decision points (score, offer, recommendations, assistant).
- **Minimisation.** Retraining without country and the UK flag gives PR-AUC 0.8732 against 0.8726 and recall 55.9% against 54.8%: no loss, so they will be removed at the next retraining.
- **Retention.** Prediction rows are deleted after 120 days (90-day label delay plus 30), implemented as a purge at start-up and unit-tested. Open: a scheduled purge, erasure by request ID, a named privacy contact and consent capture.
- **Model and data cards** (MODEL_CARD.md, DATA_CARD.md) state intended and out-of-scope use, metrics, subgroup results, provenance and the licence check still to be done against the UCI page before any redistribution.

12. Limitations, next steps and recommendation

- Public dataset, one chronological held-out slice of 788 customers, which was also used to sweep the profit threshold (Section 4.4 shows the result does not change when the threshold is chosen on the calibration slice): the held-out PR-AUC recomputed today on the same split is 0.8726 against 0.8753 in the metadata, a difference I did not investigate. Real accuracy needs labelled live predictions (about 24 Jan 2027 at the earliest).
- The profit case rests on three unverified assumptions and fails in 5 of 27 combined scenarios (Section 4.5). The first task of a pilot is to measure the real coupon recovery rate with a control group.
- The model does not separate retained from churned customers among new and low-spend customers. More tuning will not fix it; new information will.
- The monitoring log is container-local (SQLite) and resets on every deploy; PostgreSQL is the production path and is proven in Docker Compose.
- Assistant quality with real LLMs and multilingual embeddings, the SageMaker tuning job, a concurrent load test and a real cost comparison were not done. Base images are not digest-pinned and web dependencies use ranges.
- **Next:** remove country at retraining; add per-segment offer rates to the monitoring page; move the log to PostgreSQL with a scheduled purge; collect features that separate low-spend and new customers; run the assistant evaluation with a real model.
- **Recommendation:** a three-month pilot on real data with a control group, staff approval of every offer, and three end tests: profit above the blanket coupon, no group systematically left out, no red alert for a month.

13. Use of AI tools, and where they had to be corrected

I used Claude (Anthropic) as a coding and writing assistant throughout: for code scaffolding, CI and deployment diagnosis, drafting the reports, and for the analysis scripts in D5. I ran the training, the tuning, the deployments and the live measurements myself, and the numbers in this report come from scripts in the repository or from the live service. AI output was wrong several times, and the checks (tests, CI, the live service, the specification) caught it. The main cases:

What went wrong	How it was found	What changed
The artifact committed at tag v0.3 was an earlier model (trained 29 Sep, PR-AUC 0.8616, threshold 0.80), not the tuned one described in D3	Found while wiring the container in D4	Correct artifact on main; v0.3 tag left as submitted and disclosed in D4; retraining reproduces D3 within about 0.002
The weekly retrain reported a gain of 0.0018 that was noise from an unstable sort at the split boundary (about 34 customers share the same last-purchase day)	First GitHub Actions run	Deterministic split (stable sort, ID tie-break); the 0.005 margin kept; the job described as a mechanism demo
The first deploy failed at role assumption: the OIDC subject now embeds numeric owner and repository IDs; wildcards did not help	CloudTrail showed the real subject	Strict trust policy with the exact ID-based string
The D2 plan named AWS App Runner	Closed to new customers; Streamlit needs WebSockets	ECS Express Mode
The interface did not always answer in the language the user typed	Raised while testing the French assistant (pull request #6)	Interface now replies in the language of the question
The first D5 draft did not follow the course specification	Re-reading the specification	D5 rebuilt: latency by endpoint, scalability, cost, fairness with three or more mitigations, governance, cards
/explain with calibrated scores gave identical effects for four groups (isotonic ties); an integer column was silently upcast	Testing on customer 12347; a failing test	Raw forest score, grouped occlusion, float frame
A Monitoring table in an earlier D5 draft stated the amber rule wrongly; and the assistant stated that the model was trained on scikit-learn 1.6.1	Comparison with the code; the model metadata records 1.8.0	Table corrected; library versions pinned to 1.8.0
An executive-summary draft said the targeted policy beat the blanket coupon in every scenario tested	Re-running the sensitivity with all three assumptions varied together (scripts/profit_sensitivity.py)	Now states the 5 of 27 failing combinations and recommends a control group

Table 8. Corrections. Items before D5 were reported in the D3 and D4 notes; D3 also listed its own changes from D2 (22 to 28 predictors, 9 model families, Head B rules mined on pre-cutoff baskets only).

Reproducing the work. The Makefile has setup, etl, rules, spotcheck, tune, train, index, test, lint, up, down and audit targets; docker compose up --build starts the stack. The README quickstart was verified in CI by the Docker smoke test; a clean-machine run is still to be done and recorded.

Appendix A. Traceability to the course deliverables

Requirement (specification)	Where it is satisfied	Evidence in the repository
D2: architecture, data and feature design, rationale, evidence boundary	Sections 2, 3, 4.1, 5, 6	D2 report; docs/ cleaning, leakage and rule files; ARCHITECTURE.md
D3: nine model families with baselines, tuning, calibration, protected held-out test, leakage tests, artifact and metadata	Sections 2, 4	docs/head_a_*.csv and .png, mlflow_comparison.png, models/*.metadata.json, src/
D3: association rules, Apriori vs FP-Growth, sparse matrix, hit-rate@5 against a popularity baseline	Section 5	docs/head_b_*.csv and .json
D4: live API and interface, health/readiness/version, containers, CI, CD, monitoring, delayed labels	Sections 7, 8	Dockerfile, Dockerfile.ui, docker-compose.yml, .github/workflows/, app/monitoring.py, D4 note
D5: latency by endpoint with before/after, scalability, cost per 1,000 predictions and assistant queries	Section 9	scripts/perf_profile.py, scale_probe.py; reports/performance/
D5: interpretability and honesty of customer-facing explanations	Section 10	scripts/importance.py; POST /explain; reports/interpretability/
D5: subgroup fairness audit with three or more mitigations and trade-offs	Section 10	scripts/fairness_audit.py; reports/fairness/
D5: Law 25 governance, model card, data card, monitoring policy	Sections 8, 11	GOVERNANCE_LOI25.md, MODEL_CARD.md, DATA_CARD.md
D6: consolidated report with corrections; bilingual executive summary; README; deck	This report; separate files	977_D6_Chauhan_Executive_Summary.pdf, README.md, deck PDF

Table A1. Traceability.

Appendix B. Reproducing the main results

Result	Command	What to expect
All tests	python -m pytest -q	60 passed (Python 3.11, same as CI)
Lint and types	ruff check app scripts tests ui; mypy	No errors
Latency, in process	python scripts/perf_profile.py	Single prediction about 35 ms against 110 ms with n_jobs=-1 (development machine)
Latency, live	python scripts/perf_profile.py --api <API URL>	Table 6 (values vary with network and load)
Fairness audit	python scripts/fairness_audit.py	reports/fairness/*.csv and the figure
Importance	python scripts/importance.py	reports/interpretability/*.csv
Profit sensitivity	python scripts/profit_sensitivity.py	Base gain $1.06 per customer per campaign; 5 of 27 negative
Drift demonstration	python scripts/simulate_traffic.py --n 200 --drift --api <API URL>	Frequency features turn red on the monitoring page
Local stack	docker compose up --build	API on 8000, interface on 8501, PostgreSQL log

Table B1. Commands. The raw dataset is not in the repository; DATA_CARD.md gives its source.